ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)
Division II & IV

Instructor

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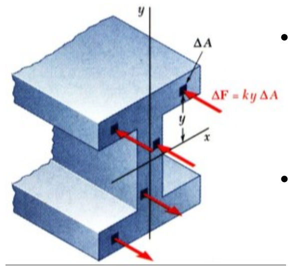
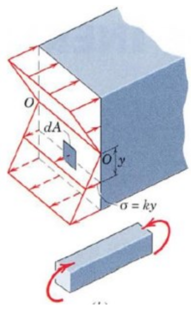
Area Moments of Inertia

- Previously considered distributed forces which were proportional to the area or volume over which they act.
  - The resultant was obtained by summing or integrating over the areas or volumes.
  - The moment of the resultant about any axis was determined by computing the first moments of the areas or volumes about that axis.
- Will now consider forces which are proportional to the area or volume over which they act but also vary linearly with distance from a given axis.
  - the magnitude of the resultant depends on the first moment of the force distribution with respect to the axis.
  - The point of application of the resultant depends on the second moment of the distribution with respect to the axis.

Area Moments of Inertia





- Consider distributed forces $\Delta\vec{F}$ whose magnitudes are proportional to the elemental areas ΔA on which they act and also vary linearly with the distance of ΔA from a given axis.
- Example: Consider a beam subjected to pure bending. Internal forces vary linearly with distance from the neutral axis which passes through the section centroid.

$$\Delta\vec{F} = ky\Delta A$$

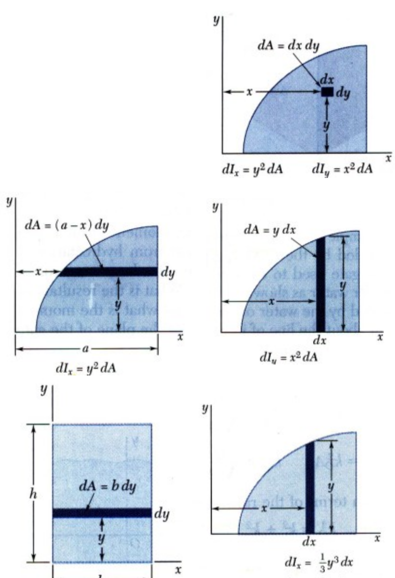
$$R = k\int y\,dA = 0 \qquad \int y\,dA = Q_x = \text{first moment}$$

$$M = k\int y^2 dA \qquad \int y^2 dA = \text{second moment}$$

First Moment of the whole section about the x-axis = $\bar{y}A$ = 0 since the centroid of the section lies on the x-axis.

Second Moment or the Moment of Inertia of the beam section about x-axis is denoted by I_x and has units of (length)^4 (never –ve)

Area Moments of Inertia by Integration



- *Second moments* or *moments of inertia* of an area with respect to the *x* and *y* axes,

$$I_x = \int y^2 dA \qquad I_y = \int x^2 dA$$

- Evaluation of the integrals is simplified by choosing dA to be a thin strip parallel to one of the coordinate axes.
- For a rectangular area,

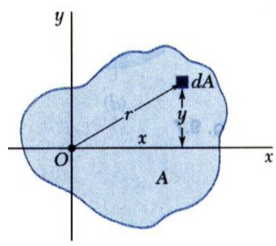
$$I_x = \int y^2 dA = \int_0^h y^2 b\,dy = \tfrac{1}{3}bh^3$$

- The formula for rectangular areas may also be applied to strips parallel to the axes,

$$dI_x = \tfrac{1}{3}y^3 dx \qquad dI_y = x^2 dA = x^2 y\,dx$$

Area Moments of Inertia

Polar Moment of Inertia



- The *polar moment of inertia* is an important parameter in problems involving torsion of cylindrical shafts and rotations of slabs.

$$J_0 = I_z = \int r^2 dA$$

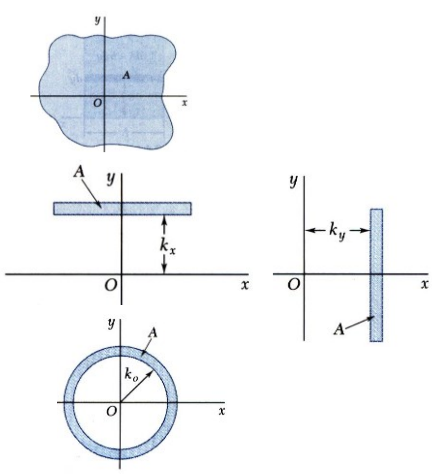
- The polar moment of inertia is related to the rectangular moments of inertia,

$$J_0 = I_z = \int r^2 dA = \int \left(x^2 + y^2\right) dA = \int x^2 dA + \int y^2 dA$$
$$= I_y + I_x$$

Moment of Inertia of an area is purely a mathematical property of the area and in itself has no physical significance.

Area Moments of Inertia

Radius of Gyration of an Area



- Consider area A with moment of inertia I_x. Imagine that the area is concentrated in a thin strip parallel to the x axis with equivalent I_x.

$$I_x = k_x^2 A \qquad k_x = \sqrt{\frac{I_x}{A}}$$

$k_x =$ *radius of gyration* with respect to the x axis

- Similarly,

$$I_y = k_y^2 A \qquad k_y = \sqrt{\frac{I_y}{A}}$$

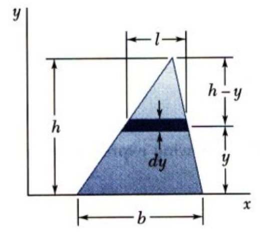
$$J_O = I_z = k_O^2 A = k_z^2 A \qquad k_O = k_z = \sqrt{\frac{J_O}{A}}$$

$$k_O^2 = k_z^2 = k_x^2 + k_y^2$$

Radius of Gyration, k is a measure of distribution of area from a reference axis
Radius of Gyration is different from centroidal distances

Area Moments of Inertia

Example: Determine the moment of inertia of a triangle with respect to its base.



SOLUTION:

- A differential strip parallel to the x axis is chosen for dA.

$$dI_x = y^2 dA \qquad dA = l\,dy$$

- For similar triangles,

$$\frac{l}{b} = \frac{h-y}{h} \qquad l = b\frac{h-y}{h} \qquad dA = b\frac{h-y}{h}dy$$

- Integrating dI_x from $y = 0$ to $y = h$,

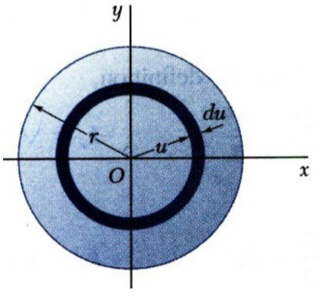
$$I_x = \int y^2 dA = \int_0^h y^2 b\frac{h-y}{h}dy = \frac{b}{h}\int_0^h \left(hy^2 - y^3\right)dy$$

$$= \frac{b}{h}\left[h\frac{y^3}{3} - \frac{y^4}{4}\right]_0^h \qquad \boxed{I_x = \frac{bh^3}{12}}$$

Area Moments of Inertia

Example: a) Determine the centroidal polar moment of inertia of a circular area by direct integration.
b) Using the result of part a, determine the moment of inertia of a circular area with respect to a diameter.

SOLUTION:



- An annular differential area element is chosen,

$$dJ_O = u^2 dA \qquad dA = 2\pi u\,du$$

$$J_O = \int dJ_O = \int_0^r u^2 (2\pi u\,du) = 2\pi\int_0^r u^3 du$$

$$\boxed{J_O = \frac{\pi}{2}r^4}$$

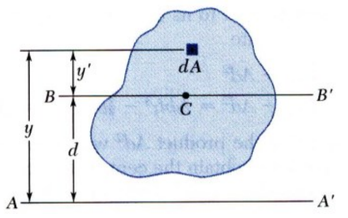
- From symmetry, $I_x = I_y$,

$$J_O = I_x + I_y = 2I_x \qquad \frac{\pi}{2}r^4 = 2I_x$$

$$\boxed{I_{diameter} = I_x = \frac{\pi}{4}r^4}$$

Area Moments of Inertia

Parallel Axis Theorem



- Consider moment of inertia I of an area A with respect to the axis AA'

$$I = \int y^2 dA$$

- The axis BB' passes through the area centroid and is called a *centroidal axis.*

$$I = \int y^2 dA = \int (y' + d)^2 dA$$
$$= \int y'^2 dA + 2d \int y' dA + d^2 \int dA$$

Parallel Axis theorem:
MI @ any axis =
MI @ centroidal axis + Ad^2

The two axes should be parallel to each other.

- Second term = 0 since centroid lies on BB' ($\int y' dA = y_c A$, and $y_c = 0$

$$I = \bar{I} + Ad^2 \quad \textit{Parallel Axis theorem}$$

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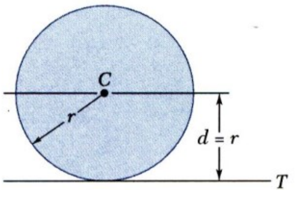
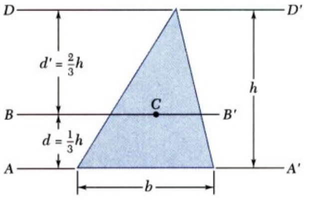
Area Moments of Inertia

Parallel Axis Theorem



- Moment of inertia I_T of a circular area with respect to a tangent to the circle,

$$I_T = \bar{I} + Ad^2 = \tfrac{1}{4}\pi r^4 + \left(\pi r^2\right) r^2$$
$$= \tfrac{5}{4}\pi r^4$$



- Moment of inertia of a triangle with respect to a centroidal axis,

$$I_{AA'} = \bar{I}_{BB'} + Ad^2$$
$$I_{BB'} = I_{AA'} - Ad^2 = \tfrac{1}{12} bh^3 - \tfrac{1}{2} bh\left(\tfrac{1}{3} h\right)^2$$
$$= \tfrac{1}{36} bh^3$$

- The moment of inertia of a composite area A about a given axis is obtained by adding the moments of inertia of the component areas A_1, A_2, A_3, ... , with respect to the same axis.

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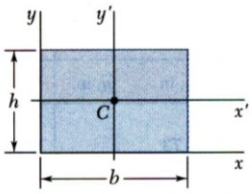
Area Moments of Inertia: Standard MIs



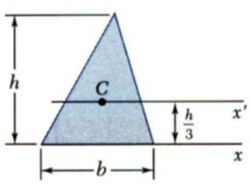
$\bar{I}_{x'} = \frac{1}{12}bh^3$

$\bar{I}_{y'} = \frac{1}{12}b^3h$

$I_x = \frac{1}{3}bh^3$

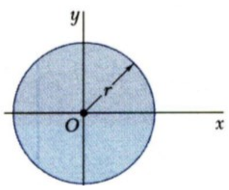
$I_y = \frac{1}{3}b^3h$

$J_C = \frac{1}{12}bh(b^2 + h^2)$



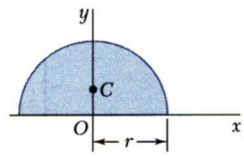
$\bar{I}_{x'} = \frac{1}{36}bh^3$

$I_x = \frac{1}{12}bh^3$



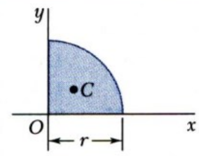
$\bar{I}_x = \bar{I}_y = \frac{1}{4}\pi r^4$

$J_O = \frac{1}{2}\pi r^4$



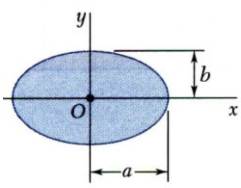
$I_x = I_y = \frac{1}{8}\pi r^4$

$J_O = \frac{1}{4}\pi r^4$



$I_x = I_y = \frac{1}{16}\pi r^4$

$J_O = \frac{1}{8}\pi r^4$



$\bar{I}_x = \frac{1}{4}\pi ab^3$

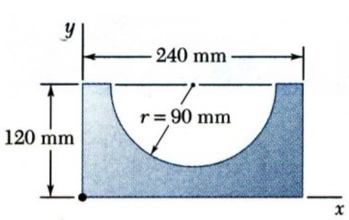
$\bar{I}_y = \frac{1}{4}\pi a^3b$

$J_O = \frac{1}{4}\pi ab(a^2 + b^2)$

Area Moments of Inertia

Example:

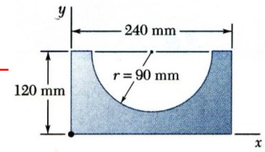
Determine the moment of inertia of the shaded area with respect to the *x* axis.



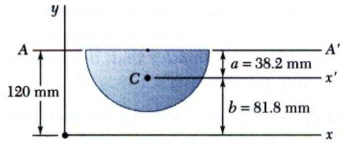
<u>SOLUTION</u>:

- Compute the moments of inertia of the bounding rectangle and half-circle with respect to the *x* axis.
- The moment of inertia of the shaded area is obtained by subtracting the moment of inertia of the half-circle from the moment of inertia of the rectangle.

Area Moments of Inertia



Example: Solution



$$a = \frac{4r}{3\pi} = \frac{(4)(90)}{3\pi} = 38.2\ mm$$

$$b = 120 - a = 81.8\ mm$$

$$A = \tfrac{1}{2}\pi r^2 = \tfrac{1}{2}\pi(90)^2$$

$$= 12.72 \times 10^3\ mm^2$$

SOLUTION:

- Compute the moments of inertia of the bounding rectangle and half-circle with respect to the x axis.

Rectangle:

$$I_x = \tfrac{1}{3}bh^3 = \tfrac{1}{3}(240)(120)^3 = 138.2 \times 10^6\ mm^4$$

Half-circle:

moment of inertia with respect to AA',

$$I_{AA'} = \tfrac{1}{8}\pi r^4 = \tfrac{1}{8}\pi(90)^4 = 25.76 \times 10^6\ \text{mm}^4$$

Moment of inertia with respect to x',

$$\bar{I}_{x'} = I_{AA'} - Aa^2 = \left(25.76 \times 10^6\right) - \left(12.72 \times 10^3\right)(38.2)^2$$

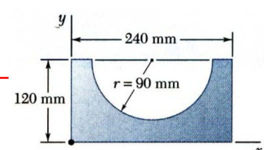
$$= 7.20 \times 10^6\ mm^4$$

moment of inertia with respect to x,

$$I_x = \bar{I}_{x'} + Ab^2 = 7.20 \times 10^6 + \left(12.72 \times 10^3\right)(81.8)^2$$

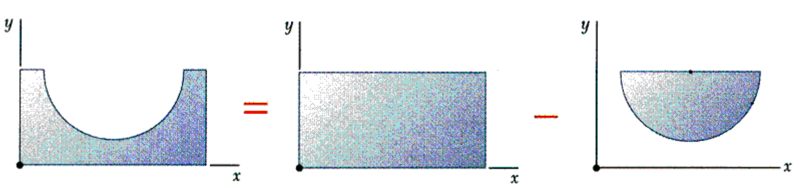
$$= 92.3 \times 10^6\ mm^4$$

Area Moments of Inertia



Example: Solution

- The moment of inertia of the shaded area is obtained by subtracting the moment of inertia of the half-circle from the moment of inertia of the rectangle.

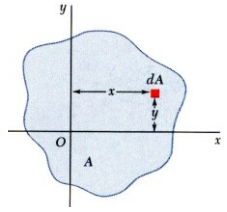


$$I_x \quad = \quad 138.2 \times 10^6\ \text{mm}^4 \quad - \quad 92.3 \times 10^6\ \text{mm}^4$$

$$\boxed{I_x = 45.9 \times 10^6\ \text{mm}^4}$$

Area Moments of Inertia

Products of Inertia: for problems involving unsymmetrical cross-sections and in calculation of MI about rotated axes. It may be +ve, -ve, or zero

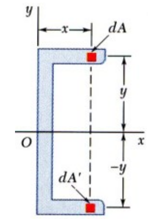


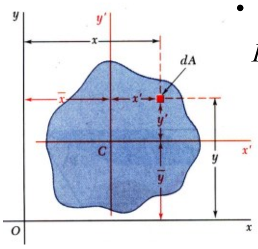
- Product of Inertia of area A w.r.t. x-y axes:

$$I_{xy} = \int xy\, dA$$

x and y are the coordinates of the element of area $dA=xy$

- When the x axis, the y axis, or both are an axis of symmetry, the product of inertia is zero.

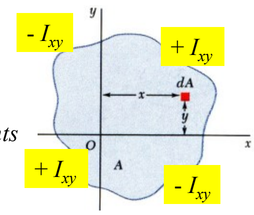




- Parallel axis theorem for products of inertia:

$$I_{xy} = \bar{I}_{xy} + \bar{x}\bar{y}A$$

Quadrants



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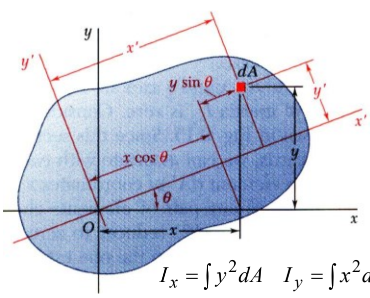
Area Moments of Inertia

Rotation of Axes

Product of inertia is useful in calculating MI @ inclined axes.

→ Determination of axes about which the MI is a maximum and a minimum



$$I_x = \int y^2 dA \quad I_y = \int x^2 dA$$

$$I_{xy} = \int xy\, dA$$

Moments and product of inertia w.r.t. new axes x' and y' ?

Note: $x' = x\cos\theta + y\sin\theta$

$y' = y\cos\theta - x\sin\theta$

$$I_{x'} = \int y'^2 dA = \int (y\cos\theta - x\sin\theta)^2 dA$$

$$I_{y'} = \int x'^2 dA = \int (x\cos\theta + y\sin\theta)^2 dA$$

$$I_{x'y'} = \int x'y' dA = \int (x\cos\theta + y\sin\theta)(y\cos\theta - x\sin\theta) dA$$

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2}$$

$$\sin\theta\cos\theta = 1/2\sin 2\theta \qquad \cos^2\theta - \sin^2\theta = \cos 2\theta$$

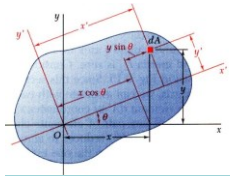
$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}\cos 2\theta - I_{xy}\sin 2\theta$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}\cos 2\theta + I_{xy}\sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2}\sin 2\theta + I_{xy}\cos 2\theta$$

Area Moments of Inertia

Rotation of Axes



$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}\cos 2\theta - I_{xy}\sin 2\theta$$
$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}\cos 2\theta + I_{xy}\sin 2\theta$$
$$I_{x'y'} = \frac{I_x - I_y}{2}\sin 2\theta + I_{xy}\cos 2\theta$$

Adding first two eqns:
$I_{x'} + I_{y'} = I_x + I_y = I_z$ → The Polar MI @ O

Angle which makes $I_{x'}$ and $I_{y'}$ either max or min can be found by setting the derivative of either $I_{x'}$ or $I_{y'}$ w.r.t. θ equal to zero:

$$\frac{dI_{x'}}{d\theta} = \left(I_y - I_x\right)\sin 2\theta - 2I_{xy}\cos 2\theta = 0$$

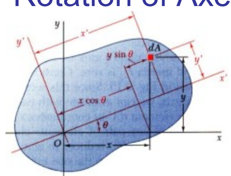
Denoting this critical angle by α

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

→ two values of 2α which differ by π since $\tan 2\alpha = \tan(2\alpha + \pi)$
→ two solutions for α will differ by $\pi/2$
→ one value of α will define the axis of maximum MI and the other defines the axis of minimum MI
→ These two rectangular axes are called the principal axes of inertia

Area Moments of Inertia

Rotation of Axes



$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}\cos 2\theta - I_{xy}\sin 2\theta$$
$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}\cos 2\theta + I_{xy}\sin 2\theta$$
$$I_{x'y'} = \frac{I_x - I_y}{2}\sin 2\theta + I_{xy}\cos 2\theta$$

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x} \Rightarrow \sin 2\alpha = \cos 2\alpha \frac{2I_{xy}}{I_y - I_x}$$

Substituting in the third eqn for critical value of 2θ: $I_{x'y'} = 0$

→ Product of Inertia $I_{x'y'}$ is zero for the Principal Axes of inertia

Substituting $\sin 2\alpha$ and $\cos 2\alpha$ in first two eqns for **Principal Moments of Inertia**:

$$I_{max} = \frac{I_x + I_y}{2} + \frac{1}{2}\sqrt{\left(I_x - I_y\right)^2 + 4I_{xy}^2}$$
$$I_{min} = \frac{I_x + I_y}{2} - \frac{1}{2}\sqrt{\left(I_x - I_y\right)^2 + 4I_{xy}^2}$$
$$I_{xy@\alpha} = 0$$

Area Moments of Inertia

Mohr's Circle of Inertia: Following relations can be represented graphically by a diagram called Mohr's Circle

For given values of I_x, I_y, & I_{xy}, corresponding values of $I_{x'}$, $I_{y'}$, & $I_{x'y'}$ may be determined from the diagram for any desired angle θ.

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}\cos 2\theta - I_{xy}\sin 2\theta$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}\cos 2\theta + I_{xy}\sin 2\theta$$

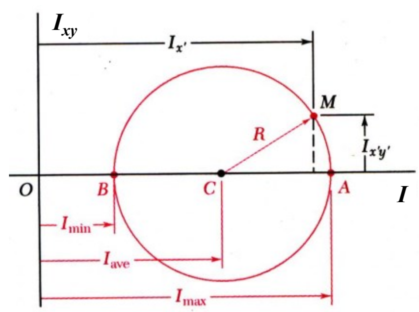
$$I_{x'y'} = \frac{I_x - I_y}{2}\sin 2\theta + I_{xy}\cos 2\theta$$

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

$$I_{\max} = \frac{I_x + I_y}{2} + \frac{1}{2}\sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{\min} = \frac{I_x + I_y}{2} - \frac{1}{2}\sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{xy@\alpha} = 0$$



$$(I_{x'} - I_{ave})^2 + I_{x'y'}^2 = R^2$$

$$I_{ave} = \frac{I_x + I_y}{2} \qquad R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

- At the points A and B, $I_{x'y'} = 0$ and $I_{x'}$ is a maximum and minimum, respectively.

$$I_{\max,\min} = I_{ave} \pm R$$

Area Moments of Inertia

Mohr's Circle of Inertia: Construction

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2}\cos 2\theta - I_{xy}\sin 2\theta$$

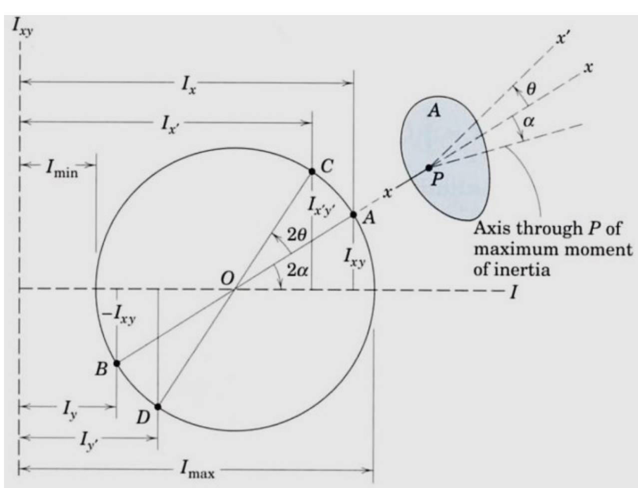
$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2}\cos 2\theta + I_{xy}\sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2}\sin 2\theta + I_{xy}\cos 2\theta$$

$$I_{\max} = \frac{I_x + I_y}{2} + \frac{1}{2}\sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{\min} = \frac{I_x + I_y}{2} - \frac{1}{2}\sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

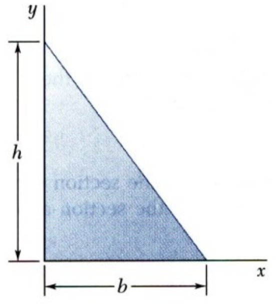
$$I_{xy@\alpha} = 0$$



Choose horz axis → MI
Choose vert axis → PI
Point A – known $\{I_x, I_{xy}\}$
Point B – known $\{I_y, -I_{xy}\}$
Circle with dia AB
Angle α for Area
→ Angle 2α to horz (same sense) → I_{max}, I_{min}
Angle x to $x' = \theta$
→ Angle OA to OC = 2θ
→ Same sense
Point C → $I_{x'}$, $I_{x'y'}$
Point D → $I_{y'}$

Area Moments of Inertia

Example: Product of Inertia



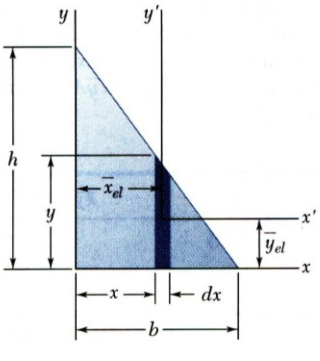
Determine the product of inertia of the right triangle *(a)* with respect to the x and y axes and *(b)* with respect to centroidal axes parallel to the x and y axes.

SOLUTION:

- Determine the product of inertia using direct integration on vertical differential area strips
- Apply the parallel axis theorem to evaluate the product of inertia with respect to the centroidal axes.

Area Moments of Inertia

Examples



SOLUTION:

- Determine the product of inertia using direct integration on vertical differential area strips

$$y = h\left(1-\frac{x}{b}\right) \qquad dA = y\,dx = h\left(1-\frac{x}{b}\right)dx$$

$$\bar{x}_{el} = x \qquad \bar{y}_{el} = \tfrac{1}{2}y = \tfrac{1}{2}h\left(1-\frac{x}{b}\right)$$

Integrating dI_x from $x = 0$ to $x = b$,

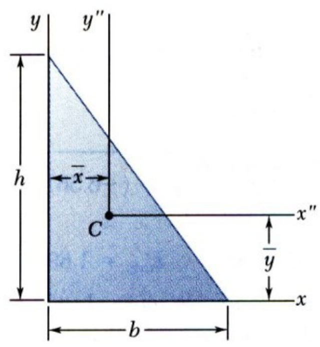
$$I_{xy} = \int dI_{xy} = \int \bar{x}_{el}\bar{y}_{el}\,dA = \int_0^b x\left(\tfrac{1}{2}\right)h^2\left(1-\frac{x}{b}\right)^2 dx$$

$$= h^2\int_0^b\left(\frac{x}{2}-\frac{x^2}{b}+\frac{x^3}{2b^2}\right)dx = h^2\left[\frac{x^2}{4}-\frac{x^3}{3b}+\frac{x^4}{8b^2}\right]_0^b$$

$$\boxed{I_{xy} = \tfrac{1}{24}b^2h^2}$$

Area Moments of Inertia

Examples



SOLUTION

- Apply the parallel axis theorem to evaluate the product of inertia with respect to the centroidal axes.

$$\bar{x} = \tfrac{1}{3}b \qquad \bar{y} = \tfrac{1}{3}h$$

With the results from part *a,*

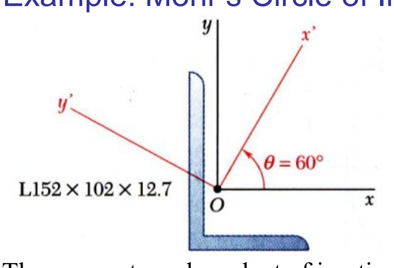
$$I_{xy} = \bar{I}_{x''y''} + \bar{x}\bar{y}A$$

$$\bar{I}_{x''y''} = \tfrac{1}{24}b^2h^2 - \left(\tfrac{1}{3}b\right)\left(\tfrac{1}{3}h\right)\left(\tfrac{1}{2}bh\right)$$

$$\boxed{\bar{I}_{x''y''} = -\tfrac{1}{72}b^2h^2}$$

Area Moments of Inertia

Example: Mohr's Circle of Inertia



The moments and product of inertia with respect to the x and y axes are I_x = 7.24x106 mm^4, I_y = 2.61x106 mm^4, and I_{xy} = -2.54x10^6 mm^4.

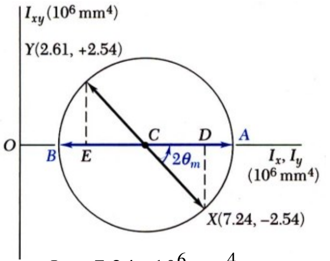
Using Mohr's circle, determine *(a)* the principal axes about O, *(b)* the values of the principal moments about O, and *(c)* the values of the moments and product of inertia about the x' and y' axes

SOLUTION:

- Plot the points (I_x, I_{xy}) and (I_y,-I_{xy}). Construct Mohr's circle based on the circle diameter between the points.
- Based on the circle, determine the orientation of the principal axes and the principal moments of inertia.
- Based on the circle, evaluate the moments and product of inertia with respect to the $x'y'$ axes.

Area Moments of Inertia

Example: Mohr's Circle of Inertia



$I_x = 7.24\times10^6\,\text{mm}^4$

$I_y = 2.61\times10^6\,\text{mm}^4$

$I_{xy} = -2.54\times10^6\,\text{mm}^4$

SOLUTION:

- Plot the points (I_x, I_{xy}) and (I_y,-I_{xy}). Construct Mohr's circle based on the circle diameter between the points.

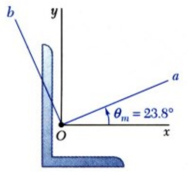
$$OC = I_{ave} = \tfrac{1}{2}\left(I_x + I_y\right) = 4.925\times10^6\,\text{mm}^4$$

$$CD = \tfrac{1}{2}\left(I_x - I_y\right) = 2.315\times10^6\,\text{mm}^4$$

$$R = \sqrt{(CD)^2 + (DX)^2} = 3.437\times10^6\,\text{mm}^4$$

- Based on the circle, determine the orientation of the principal axes and the principal moments of inertia.

$$\tan 2\theta_m = \frac{DX}{CD} = 1.097 \quad 2\theta_m = 47.6^\circ \quad \boxed{\theta_m = 23.8^\circ}$$

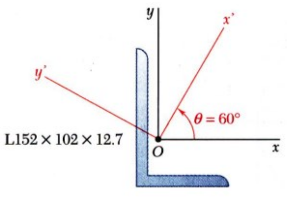


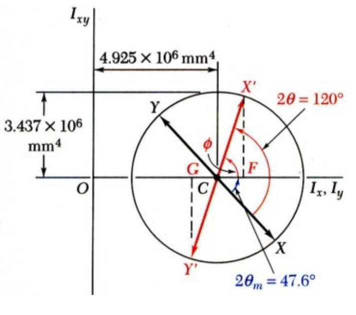
Area Moments of Inertia

$$OC = I_{ave} = 4.925\times10^6\,\text{mm}^4$$

$$R = 3.437\times10^6\,\text{mm}^4$$

Example: Mohr's Circle of Inertia





- Based on the circle, evaluate the moments and product of inertia with respect to the *x'y'* axes.

The points *X'* and *Y'* corresponding to the *x'* and *y'* axes are obtained by rotating *CX* and *CY* counterclockwise through an angle θ= 2(60°) = 120°. The angle that *CX'* forms with the horz is ϕ = 120° - 47.6° = 72.4°.

$$I_{x'} = OF = OC + CX'\cos\varphi = I_{ave} + R\cos 72.4^o$$

$$\boxed{I_{x'} = 5.96\times10^6\,\text{mm}^4}$$

$$I_{y'} = OG = OC - CY'\cos\varphi = I_{ave} - R\cos 72.4^o$$

$$\boxed{I_{y'} = 3.89\times10^6\,\text{mm}^4}$$

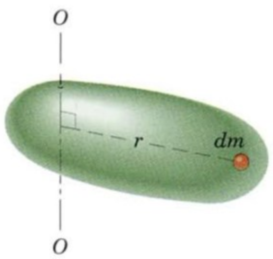
$$I_{x'y'} = FX' = CY'\sin\varphi = R\sin 72.4^o$$

$$\boxed{I_{x'y'} = 3.28\times10^6\,\text{mm}^4}$$

Mass Moment of Inertia

Mass Moments of Inertia (I): Important in Rigid Body Dynamics

- I is a measure of distribution of mass of a rigid body w.r.t. the axis in question (constant property for that axis).
- Units are (mass)(length)^2 → kg.m^2



Consider a three dimensional body of mass *m*
Mass moment of inertia of this body about axis *O-O*:

$$I = \int r^2\, dm$$

Integration is over the entire body.

r = perpendicular distance of the mass element dm from the axis *O-O*

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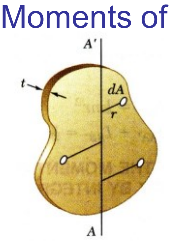
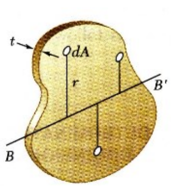
Mass Moment of Inertia

Moments of Inertia of Thin Plates



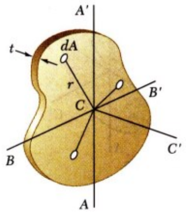
- For a thin plate of uniform thickness t and homogeneous material of density ρ, the mass moment of inertia with respect to axis AA' contained in the plate is

$$I_{AA'} = \int r^2 dm = \rho t \int r^2 dA$$
$$= \rho t\, I_{AA',area}$$



- Similarly, for perpendicular axis BB' which is also contained in the plate,

$$I_{BB'} = \rho t\, I_{BB',area}$$

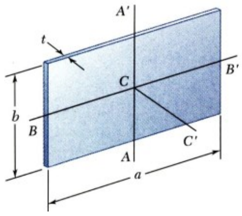


- For the axis CC' which is perpendicular to the plate,

$$I_{CC'} = \rho t\, J_{C,area} = \rho t \left(I_{AA',area} + I_{BB',area}\right)$$
$$= I_{AA'} + I_{BB'}$$

Mass Moment of Inertia

Moments of Inertia of Thin Plates

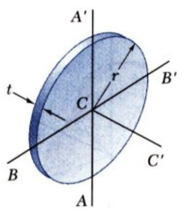


- For the principal centroidal axes on a rectangular plate,

$$I_{AA'} = \rho t\, I_{AA',area} = \rho t\left(\tfrac{1}{12} a^3 b\right) = \tfrac{1}{12} m a^2$$

$$I_{BB'} = \rho t\, I_{BB',area} = \rho t\left(\tfrac{1}{12} a b^3\right) = \tfrac{1}{12} m b^2$$

$$I_{CC'} = I_{AA',mass} + I_{BB',mass} = \tfrac{1}{12} m\left(a^2 + b^2\right)$$

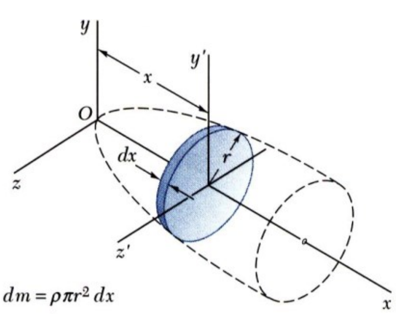


- For centroidal axes on a circular plate,

$$I_{AA'} = I_{BB'} = \rho t\, I_{AA',area} = \rho t\left(\tfrac{1}{4}\pi r^4\right) = \tfrac{1}{4} m r^2$$

Mass Moment of Inertia

Moments of Inertia of a 3D Body by Integration



$$dm = \rho \pi r^2\, dx$$

$$dI_x = \tfrac{1}{2} r^2\, dm$$

$$dI_y = dI_{y'} + x^2\, dm = \left(\tfrac{1}{4} r^2 + x^2\right) dm$$

$$dI_z = dI_{z'} + x^2\, dm = \left(\tfrac{1}{4} r^2 + x^2\right) dm$$

- Moment of inertia of a homogeneous body is obtained from double or triple integrations of the form

$$I = \rho \int r^2\, dV$$

- For bodies with two planes of symmetry, the moment of inertia may be obtained from a single integration by choosing thin slabs perpendicular to the planes of symmetry for *dm*.

- The moment of inertia with respect to a particular axis for a composite body may be obtained by adding the moments of inertia with respect to the same axis of the components.

Mass Moment of Inertia

Plane Kinetics of Rigid Bodies: Force, Mass, & Acceleration

- Application to three cases of rigid body motion:

-Translation, Fixed Axis Rotation, and General Plane Motion

Fixed Axis Rotation: Mass moment of inertia will be extensively used.

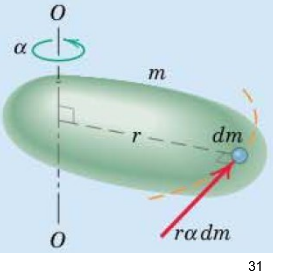
Mass Moments of Inertia

- Required in the analysis of any body that has rotational acceleration @ an axis
- Mass *m* of a body is a measure of resistance to translational acceleration
- Area moment of inertia is a measure of the distribution of area @ the axis
- Mass Moment of Inertia *I* is a measure of resistance to rotational acceleration of the body

Mass moment of inertia of the body @ *O-O*:

$$I = \int r^2\, dm \qquad I = \Sigma r_i^{\,2} m_i \qquad I = \rho \int r^2\, dV$$

If mass density ρ is constant throughout the body



Units of Mass moment of inertia: kg m^2

Mass Moment of Inertia

Mass Moments of Inertia

Radius of Gyration (*k*)

- about an axis for which *I* is defined: $k = \sqrt{\frac{I}{m}}$ or $I = k^2 m$

Parallel Axis Theorem (Transfer of Axes)

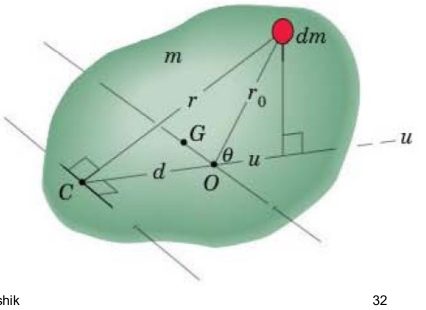
- Mass moment of inertia about any axis parallel to the axis passing through mass center *G*:

$$I = \bar{I} + md^2$$

- Radius of gyration @ an axis through *C*

$$k^2 = \bar{k}^2 + d^2$$

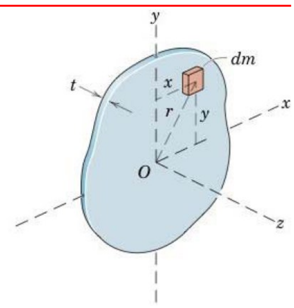
$\bar{I}$ and $\bar{k}$ are the values @ an axis through mass center



Mass Moment of Inertia

Mass Moments of Inertia

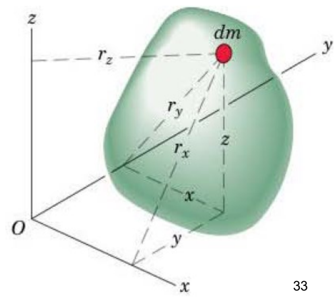
Plane Motion:
Mass MI of the plate (with motion in *x-y* plane) @ *z*-axis through *O*:



$$I_O = \int r^2\, dm \qquad I_O = \Sigma r_i^2 m_i$$

3-D Motion:

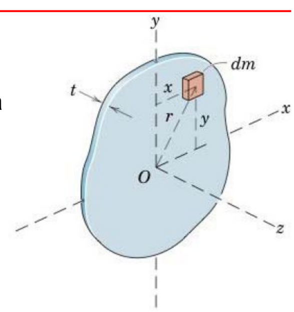
$$I_{xx} = \int r_x^2\, dm = \int (y^2 + z^2)\, dm$$
$$I_{yy} = \int r_y^2\, dm = \int (z^2 + x^2)\, dm$$
$$I_{zz} = \int r_z^2\, dm = \int (x^2 + y^2)\, dm$$



Mass Moment of Inertia

Mass Moments of Inertia

Thin Plates
Relationship between mass moments of inertia and area moments of inertia exists in case of **flat plates**.



t = constant thickness of the plate,
ρ = constant mass density of the plate

Mass MI I_{zz} of the plate @ *z*-axis normal to the plate:

$$I_{zz} = \int r^2\, dm = \rho t \int r^2\, dA = \rho t I_z$$

Mass MI @ *z*-axis = mass per unit area (*ρt*) x Polar MI of the plate area @ *z*-axis

If *t* is small compared to the dimensions of the plate in its plane:
Mass MI I_{xx} and I_{yy} of the plate @ *x*- and *y*-axes are closely approximated by:

$$I_{xx} = \int y^2\, dm = \rho t \int y^2\, dA = \rho t I_x$$
$$I_{yy} = \int x^2\, dm = \rho t \int x^2\, dA = \rho t I_y$$

For thin plates

$$I_{zz} = I_{xx} + I_{yy}$$

For thin plates

Mass Moment of Inertia

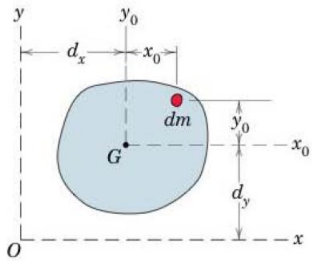
Mass Moments of Inertia

Products of Inertia: used in the expression for angular momentum of rigid bodies under 3-D motion

$$I_{xy} = I_{yx} = \int xy\, dm$$
$$I_{xz} = I_{zx} = \int xz\, dm$$
$$I_{yz} = I_{zy} = \int yz\, dm$$

$$I_{xy} = \bar{I}_{xy} + m d_x d_y$$
$$I_{xz} = \bar{I}_{xz} + m d_x d_z$$
$$I_{yz} = \bar{I}_{yz} + m d_y d_z$$

Parallel Axis Theorem

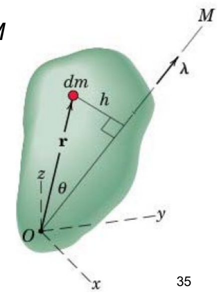


→ extremely useful while determining mass MI @ any axis *OM*
Direction cosines of *OM*: l, m, n
Unit vector along *OM*: $\boldsymbol{\lambda} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$

$$I_M = \int h^2\, dm = \int (\mathbf{r} \times \boldsymbol{\lambda})\cdot(\mathbf{r} \times \boldsymbol{\lambda})\, dm$$

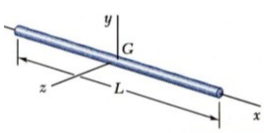
$$(\mathbf{r} \times \boldsymbol{\lambda}) = (yn - zm)\mathbf{i} + (zl - xn)\mathbf{j} + (xm - yl)\mathbf{k}$$

$$I_M = I_{xx}l^2 + I_{yy}m^2 + I_{zz}n^2 - 2I_{xy}lm - 2I_{xz}ln - 2I_{yz}mn$$

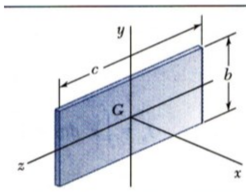


Mass Moment of Inertia

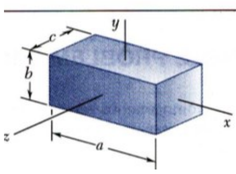
MI of some common geometric shapes



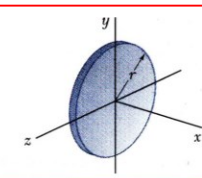
$I_y = I_z = \frac{1}{12} mL^2$



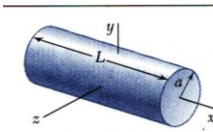
$I_x = \frac{1}{12} m(b^2 + c^2)$
$I_y = \frac{1}{12} mc^2$
$I_z = \frac{1}{12} mb^2$



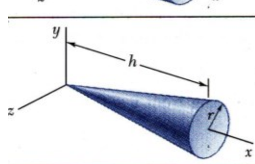
$I_x = \frac{1}{12} m(b^2 + c^2)$
$I_y = \frac{1}{12} m(c^2 + a^2)$
$I_z = \frac{1}{12} m(a^2 + b^2)$



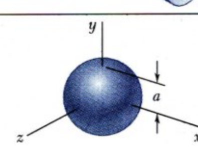
$I_x = \frac{1}{2} mr^2$
$I_y = I_z = \frac{1}{4} mr^2$



$I_x = \frac{1}{2} ma^2$
$I_y = I_z = \frac{1}{12} m(3a^2 + L^2)$



$I_x = \frac{3}{10} ma^2$
$I_y = I_z = \frac{3}{5} m(\frac{1}{4}a^2 + h^2)$



$I_x = I_y = I_z = \frac{2}{5} ma^2$

ME101: Statics

That is All!